TITLE:

Drop-off & Did Not Attend rates

Dataset Used:

nhstalkingtherapies_month_mar_2026_activity_performance.csv

SUBMITTED BY:

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DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Dataset Introduction

The NHS Talking Therapies dataset contains activity and performance statistics related to mental health services delivered across England. The dataset includes information about referrals, treatment sessions, waiting times, attendance status, completed treatments, and appointment outcomes.

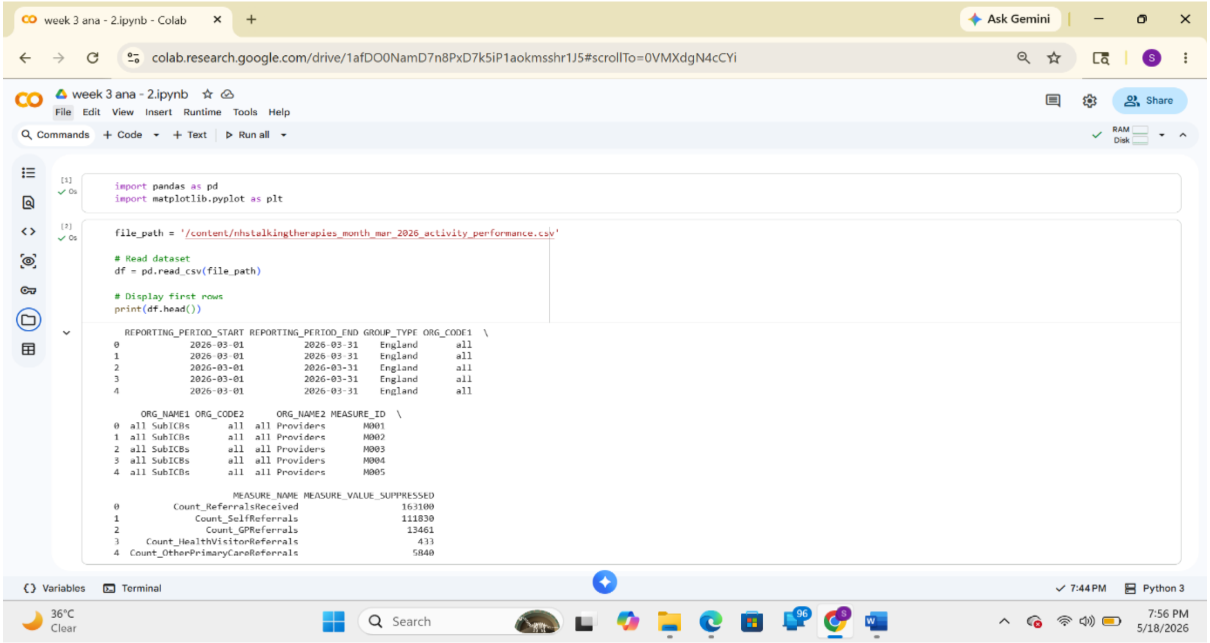
For this project, the dataset is used to:

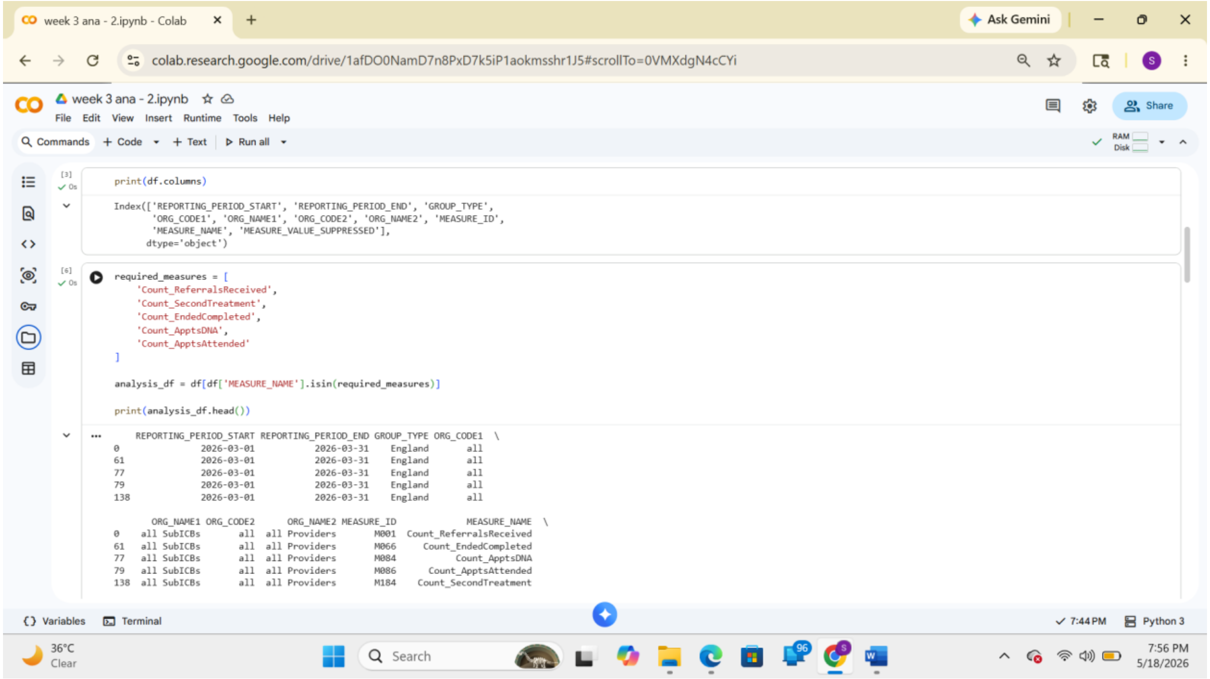
- Calculate **Did Not Attend (DNA) appointment rates**
- Analyse **treatment drop-off rates**
- Build a **funnel chart** representing patient progression through treatment stages
- Export the analysis results into a CSV file

Technologies Used

Technology	Purpose
Python	Programming language for analysis
Pandas	Data cleaning and manipulation
Matplotlib	Data visualization and funnel chart
Google Colab Notebook	Cloud environment for executing Python code

Drop-off & Did Not Attend rates





```
print(df.columns)
```

```
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'GROUP_TYPE',
       'ORG_CODE1', 'ORG_NAME1', 'ORG_CODE2', 'ORG_NAME2', 'MEASURE_ID',
       'MEASURE_NAME', 'MEASURE_VALUE_SUPPRESSED'],
      dtype='object')
```

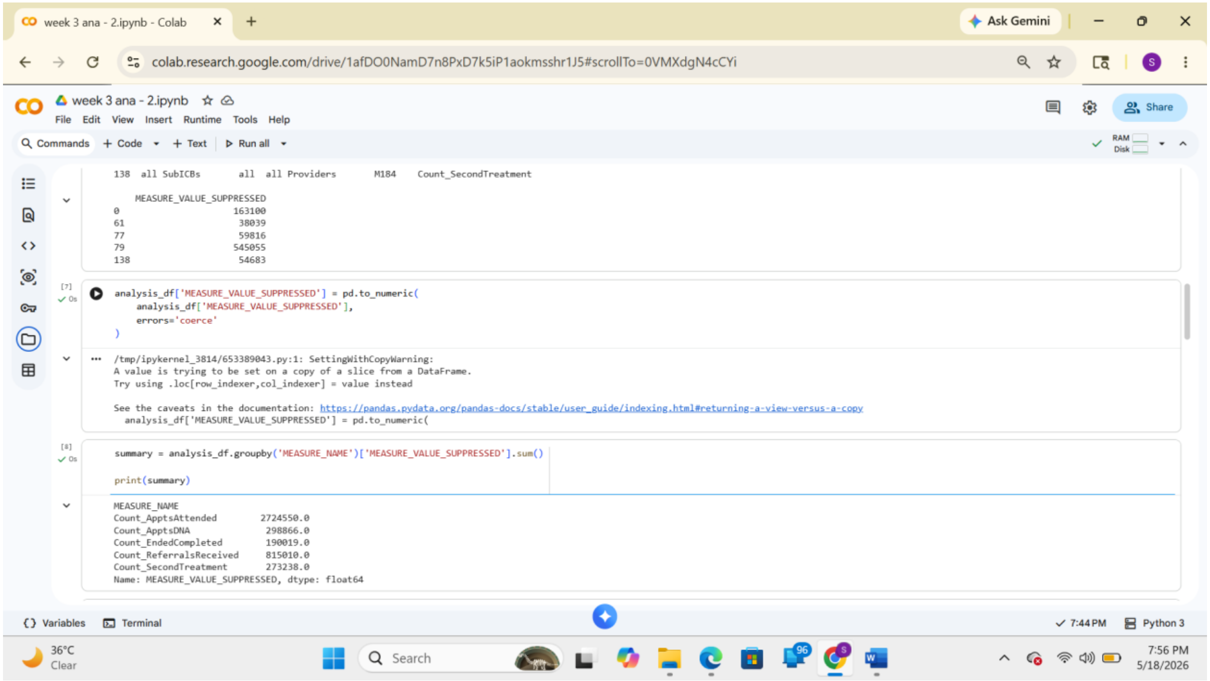
```
required_measures = [
    'Count_ReferralsReceived',
    'Count_SecondTreatment',
    'Count_EndedCompleted',
    'Count_ApptsDNA',
    'Count_ApptsAttended'
]

analysis_df = df[df['MEASURE_NAME'].isin(required_measures)]

print(analysis_df.head())
```

```
    REPORTING_PERIOD_START REPORTING_PERIOD_END GROUP_TYPE ORG_CODE1  \
0               2026-03-01           2026-03-31    England       all
61              2026-03-01           2026-03-31    England       all
77              2026-03-01           2026-03-31    England       all
79              2026-03-01           2026-03-31    England       all
138             2026-03-01           2026-03-31    England       all

      ORG_NAME1 ORG_CODE2      ORG_NAME2 MEASURE_ID             MEASURE_NAME  \
0    all SubICBs       all  all Providers       M001  Count_ReferralsReceived
61   all SubICBs       all  all Providers       M066     Count_EndedCompleted
77   all SubICBs       all  all Providers       M084           Count_ApptsDNA
79   all SubICBs       all  all Providers       M086      Count_ApptsAttended
138  all SubICBs       all  all Providers       M184    Count_SecondTreatment
```



```
138  all SubICBs       all  all Providers       M184    Count_SecondTreatment

    MEASURE_VALUE_SUPPRESSED
0                     163100
61                     38039
77                     59816
79                    545055
138                    54683
```

```
analysis_df['MEASURE_VALUE_SUPPRESSED'] = pd.to_numeric(
    analysis_df['MEASURE_VALUE_SUPPRESSED'],
    errors='coerce'
)
```

```
/tmp/ipykernel_3814/653389043.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  analysis_df['MEASURE_VALUE_SUPPRESSED'] = pd.to_numeric(
```

```
summary = analysis_df.groupby('MEASURE_NAME')['MEASURE_VALUE_SUPPRESSED'].sum()

print(summary)
```

```
MEASURE_NAME
Count_ApptsAttended        2724550.0
Count_ApptsDNA              298866.0
Count_EndedCompleted        190019.0
Count_ReferralsReceived     815010.0
Count_SecondTreatment       273238.0
Name: MEASURE_VALUE_SUPPRESSED, dtype: float64
```

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```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  analysis_df['MEASURE_VALUE_SUPPRESSED'] = pd.to_numeric(
```

[8]
```python
summary = analysis_df.groupby('MEASURE_NAME')['MEASURE_VALUE_SUPPRESSED'].sum()

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```

```
MEASURE_NAME
Count_ApptsAttended        2724550.0
Count_ApptsDNA              298866.0
Count_EndedCompleted        190019.0
Count_ReferralsReceived     815010.0
Count_SecondTreatment       273238.0
Name: MEASURE_VALUE_SUPPRESSED, dtype: float64
```

[9]
```python
referrals = summary['Count_ReferralsReceived']
second_treatment = summary['Count_SecondTreatment']
completed = summary['Count_EndedCompleted']
dna = summary['Count_ApptsDNA']
attended = summary['Count_ApptsAttended']

# Calculate rates
DNA_rate = (dna / (dna + attended)) * 100
Dropoff_rate = ((referrals - completed) / referrals) * 100

print('DNA Rate:', round(DNA_rate, 2), '%')
print('Treatment Drop-off Rate:', round(Dropoff_rate, 2), '%')
```

```
DNA Rate: 9.89 %
Treatment Drop-off Rate: 76.69 %
```

[10]
```python
funnel_df = pd.DataFrame({
```

Variables Terminal 7:44 PM Python 3

36°C Clear Search 7:57 PM 5/18/2026

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```
DNA Rate: 9.89 %
Treatment Drop-off Rate: 76.69 %
```

[10]
```python
funnel_df = pd.DataFrame({
    'Stage': [
        'Referrals Received',
        'Second Treatment',
        'Treatment Completed'
    ],
    'Count': [
        referrals,
        second_treatment,
        completed
    ]
})

print(funnel_df)
```

```
                 Stage     Count
0   Referrals Received  815010.0
1     Second Treatment  273238.0
2  Treatment Completed  190019.0
```

[11]
```python
plt.figure(figsize=(10,6))

plt.barh(
    funnel_df['Stage'],
    funnel_df['Count']
)

plt.xlabel('Patient Count')
plt.ylabel('Treatment Stage')
```

Variables Terminal 7:44 PM Python 3

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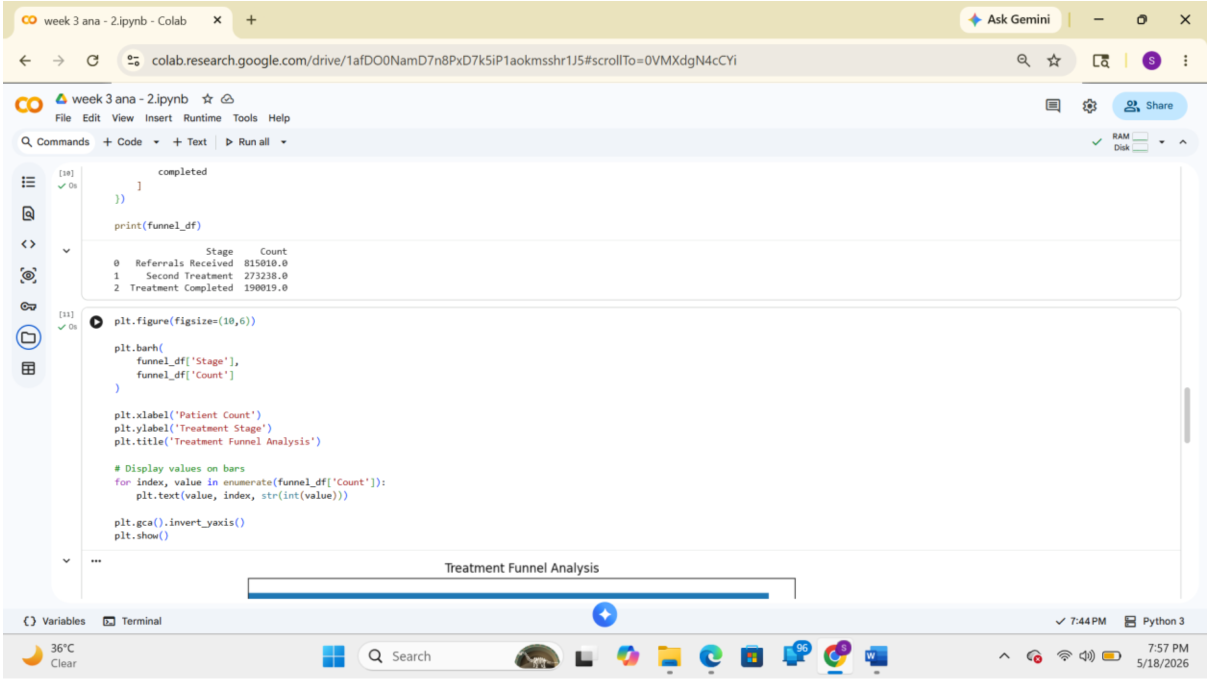
week 3 ana - 2.ipynb - Colab
Ask Gemini
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week 3 ana - 2.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
completed
]
})
print(funnel_df)
Stage Count
0 Referrals Received 815010.0
1 Second Treatment 273238.0
2 Treatment Completed 190019.0
plt.figure(figsize=(10,6))
plt.barh(
funnel_df['Stage'],
funnel_df['Count']
)
plt.xlabel('Patient Count')
plt.ylabel('Treatment Stage')
plt.title('Treatment Funnel Analysis')
# Display values on bars
for index, value in enumerate(funnel_df['Count']):
plt.text(value, index, str(int(value)))
plt.gca().invert_yaxis()
plt.show()
Treatment Funnel Analysis
Variables Terminal
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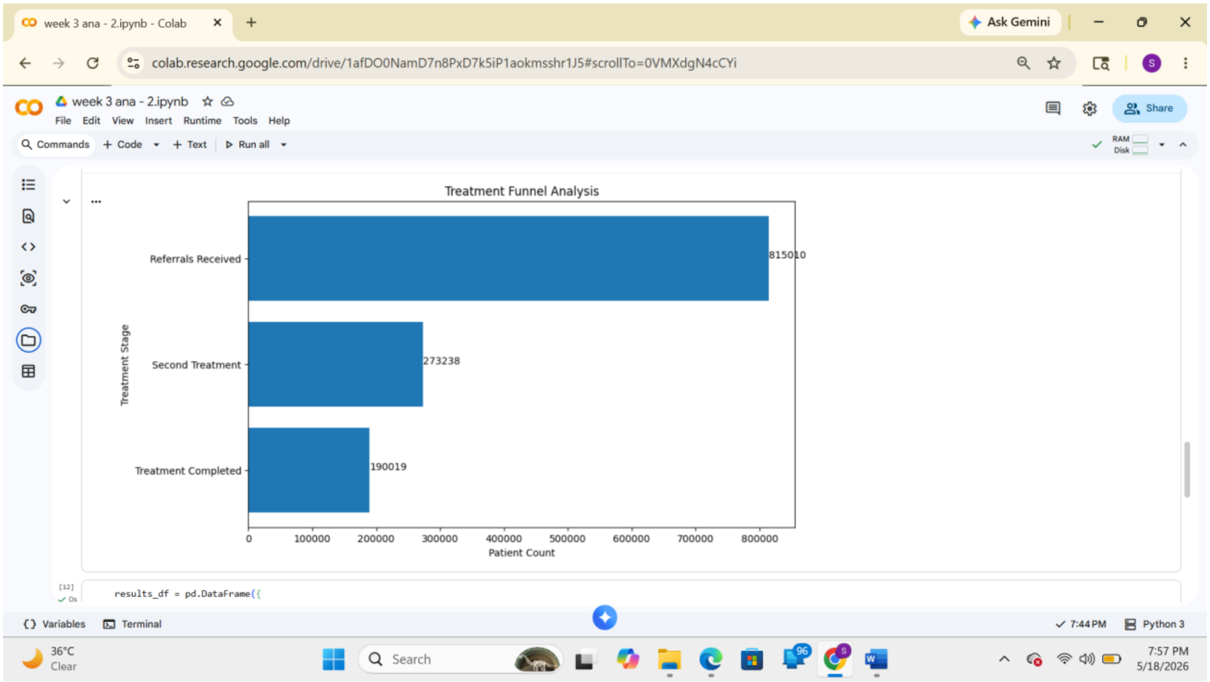
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Ask Gemini
colab.research.google.com/drive/1afDO0NamD7n8PxD7k5iP1aokmsshr1J5#scrollTo=0VMXdgN4cCYi
week 3 ana - 2.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Treatment Funnel Analysis
Referrals Received
815010
Second Treatment
273238
Treatment Completed
190019
Treatment Stage
0 100000 200000 300000 400000 500000 600000 700000 800000
Patient Count
results_df = pd.DataFrame({
Variables Terminal
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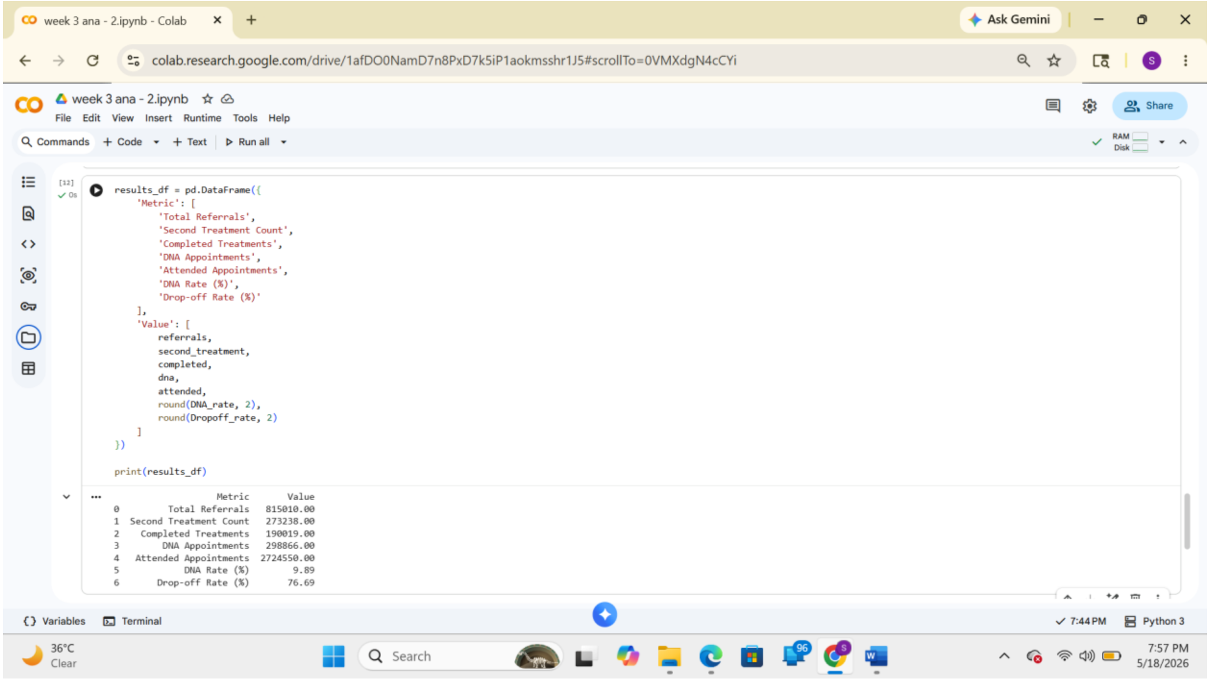
Search
7:57 PM 5/18/2026



Funnel Chart

The funnel chart shows:

1. Total referrals received
2. Patients reaching second treatment
3. Patients completing treatment

The narrowing funnel visually represents treatment drop-off.

CSV Output

The generated CSV file contains:

- Referral counts
- Treatment completion counts
- DNA appointment counts
- Attendance counts
- DNA percentage
- Treatment drop-off percentage

Conclusion

This project analyses patient attendance and treatment completion patterns using NHS Talking Therapies data.

Key Findings

- DNA rate indicates the percentage of appointments missed by patients.
- Drop-off analysis identifies how many patients discontinue treatment before completion.
- Funnel visualization helps understand patient progression through treatment stages.

Benefits of the Analysis

- Supports healthcare performance monitoring
- Helps identify service engagement issues
- Assists in improving patient retention strategies
- Enables data-driven decision making in mental health services

The project demonstrates how Python, Pandas, and Matplotlib can be used in Google Colab to perform healthcare analytics and generate meaningful visual insights.